

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673



231-773-5998 Phone
888-979-4469 Fax
www.trace-labs.com

March 23, 2026

Patrick Miller
Cadillac, City of
1121 Plett Rd.
Cadillac, MI 49601

RE: Trace Project 26C0249
Client Project NPDES AMR. - March 2026

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

A handwritten signature in black ink that reads "Timothy W. Brewer".

Tim Brewer
Project Manager
Enclosures



TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 26C0249
Client Project ID: NPDES AMR.- March 2026

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
26C0249-01	Tert. Effluent	Wastewater	TL	03/05/26 07:00	03/05/26 11:55
Sample type: Composite					
26C0249-02	Tert. Effluent	Wastewater	TL	03/05/26 08:15	03/05/26 11:55
Sample type: Grab					

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ANALYTICAL RESULTS

Trace Project ID: 26C0249
Client Project ID: NPDES AMR.- March 2026

Trace ID: 26C0249-01 Matrix: Wastewater Date Collected: 03/05/26 07:00
Sample ID: Tert. Effluent Date Received: 03/05/26 11:55

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
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METALS, TOTAL

Analysis Method: EPA 200.7 Rev. 4.4

Batch: T180080

Calcium	71 mg/L	1.0	1	03/06/26	jrc	03/12/26	drm		
Magnesium	16 mg/L	1.0	1	03/06/26	jrc	03/12/26	drm		

Analysis Method: EPA 200.8 Rev. 5.4

Batch: T180080

Antimony	0.44 ug/L	0.25	1	03/06/26	jrc	03/09/26	jma		
Chromium	0.72 ug/L	5.0	1	03/06/26	jrc	03/09/26	jma	J	
Lead	0.11 ug/L	0.55	1	03/06/26	jrc	03/09/26	jma	J	
Nickel	3.8 ug/L	5.0	1	03/06/26	jrc	03/09/26	jma	J	
Selenium	0.43 ug/L	0.50	1	03/06/26	jrc	03/09/26	jma	J	
Thallium	<0.38 ug/L	0.38	1	03/06/26	jrc	03/09/26	jma		
Zinc	35 ug/L	10	1	03/06/26	jrc	03/09/26	jma		

Analysis Method: SM 2340 B-21

Batch: [CALC]

Hardness as CaCO3	240 mg/L	2.6	1	03/06/26		03/12/26	drm	N	
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SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS

Analysis Method: EPA 625.1

Batch: T180180

N-Nitrosodimethylamine	<5.1 ug/L	5.1	1	03/10/26	kbc	03/13/26	av		
Bis(2-chloroethyl)ether	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
2-Chlorophenol	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Phenol	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
1,4-Dichlorobenzene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Bis(2-chloroisopropyl)ether	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
N-Nitrosodi-n-propylamine	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Nitrobenzene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
Isophorone	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
2-Nitrophenol	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
2,4-Dimethylphenol	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Bis(2-chloroethoxy)methane	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		

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ANALYTICAL RESULTS

Trace Project ID: 26C0249
Client Project ID: NPDES AMR.- March 2026

Trace ID: 26C0249-01 Matrix: Wastewater Date Collected: 03/05/26 07:00
Sample ID: Tert. Effluent Date Received: 03/05/26 11:55

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
1,2,4-Trichlorobenzene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
2,4-Dichlorophenol	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Naphthalene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
4-Chloro-3-methylphenol	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
2,4,6-Trichlorophenol	<1.8 ug/L	1.8	1	03/10/26	kbc	03/13/26	av		
2-Chloronaphthalene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Dimethyl phthalate	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Acenaphthylene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
2,6-Dinitrotoluene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Acenaphthene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
2,4-Dinitrotoluene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
4-Nitrophenol	<20 ug/L	20	1	03/10/26	kbc	03/13/26	av		
2,4-Dinitrophenol	<15 ug/L	15	1	03/10/26	kbc	03/13/26	av		
Diethyl phthalate	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Fluorene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
4-Chlorophenyl phenyl ether	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
4,6-Dinitro-2-methylphenol	<5.1 ug/L	5.1	1	03/10/26	kbc	03/13/26	av		
N-Nitrosodiphenylamine	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
1,2-Diphenylhydrazine	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av	N	
4-Bromophenyl phenyl ether	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Phenanthrene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
Anthracene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Di-n-butyl phthalate	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Fluoranthene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
Benzidine	<51 ug/L	51	1	03/10/26	kbc	03/13/26	av	MS07	
Pyrene	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Butyl benzyl phthalate	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Benzo (a) anthracene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
Chrysene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
3,3'-Dichlorobenzidine	<5.1 ug/L	5.1	1	03/10/26	kbc	03/13/26	av		
Bis(2-ethylhexyl)phthalate	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Di-n-octyl phthalate	<2.5 ug/L	2.5	1	03/10/26	kbc	03/13/26	av		
Benzo (b) fluoranthene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		

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ANALYTICAL RESULTS

Trace Project ID: 26C0249
Client Project ID: NPDES AMR.- March 2026

Trace ID: 26C0249-01 Matrix: Wastewater Date Collected: 03/05/26 07:00
Sample ID: Tert. Effluent Date Received: 03/05/26 11:55

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Benzo (k) fluoranthene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
Benzo (a) pyrene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
Indeno (1,2,3-cd) pyrene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
Dibenz (a,h) anthracene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
Benzo (g,h,i) perylene	<1.0 ug/L	1.0	1	03/10/26	kbc	03/13/26	av		
Surrogates:									
2-Fluorophenol	32 %	20-53	1	03/10/26	kbc	03/13/26	av		
Phenol-d5	23 %	11-40	1	03/10/26	kbc	03/13/26	av		
Nitrobenzene-d5	57 %	36-103	1	03/10/26	kbc	03/13/26	av		
2-Fluorobiphenyl	54 %	36-119	1	03/10/26	kbc	03/13/26	av	N	
2,4,6-Tribromophenol	45 %	30-105	1	03/10/26	kbc	03/13/26	av	N	
Terphenyl-d14	41 %	37-109	1	03/10/26	kbc	03/13/26	av		

PESTICIDES/PCBS

Analysis Method: EPA 612

Batch: T180228

Hexachlorobenzene	<0.0098 ug/L	0.0098	1	03/11/26	kbc	03/13/26	ahr	N	
Hexachlorobutadiene	<0.0098 ug/L	0.0098	1	03/11/26	kbc	03/13/26	ahr	N	
Hexachlorocyclopentadiene	<0.0098 ug/L	0.0098	1	03/11/26	kbc	03/13/26	ahr	N	
Hexachloroethane	<0.0098 ug/L	0.0098	1	03/11/26	kbc	03/13/26	ahr	N	
Pentachlorophenol	0.25 ug/L	1.8	1	03/11/26	kbc	03/13/26	ahr	J, N	
Surrogates:									
Tetrachloro-m-xylene	68 %	18-105	1	03/11/26	kbc	03/13/26	ahr	N	

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ANALYTICAL RESULTS

Trace Project ID: 26C0249
Client Project ID: NPDES AMR.- March 2026

Trace ID: 26C0249-02 Matrix: Wastewater Date Collected: 03/05/26 08:15
Sample ID: Tert. Effluent Date Received: 03/05/26 11:55

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
<i>Batch: T180116</i>									
Chloromethane	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Vinyl chloride	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Bromomethane	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Chloroethane	<2.0 ug/L	2.0	1	03/07/26	ats	03/07/26	ats		
Acrolein	<5.0 ug/L	5.0	1	03/07/26	ats	03/07/26	ats		
1,1-Dichloroethene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Methylene chloride	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Acrylonitrile	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
trans-1,2-Dichloroethene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
1,1-Dichloroethane	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Chloroform	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
1,1,1-Trichloroethane	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Carbon tetrachloride	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Benzene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
1,2-Dichloroethane	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Trichloroethene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
1,2-Dichloropropane	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
cis-1,2-Dichloroethene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Bromodichloromethane	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
2-Chloroethylvinyl ether	<5.0 ug/L	5.0	1	03/07/26	ats	03/07/26	ats		
cis-1,3-Dichloropropene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Toluene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
trans-1,3-Dichloropropene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
1,1,2-Trichloroethane	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Tetrachloroethene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Dibromochloromethane	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Chlorobenzene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Ethylbenzene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Bromoform	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
1,1,2,2-Tetrachloroethane	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
1,3-Dichlorobenzene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
1,4-Dichlorobenzene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		

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ANALYTICAL RESULTS

Trace Project ID: 26C0249
Client Project ID: NPDES AMR.- March 2026

Trace ID: 26C0249-02 Matrix: Wastewater Date Collected: 03/05/26 08:15
Sample ID: Tert. Effluent Date Received: 03/05/26 11:55

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
1,2-Dichlorobenzene	<1.0 ug/L	1.0	1	03/07/26	ats	03/07/26	ats		
Surrogates:									
1,2-Dichloroethane-d4	98 %	68-133	1	03/07/26	ats	03/07/26	ats		
Toluene-d8	101 %	75-120	1	03/07/26	ats	03/07/26	ats		
4-Bromofluorobenzene	104 %	69-119	1	03/07/26	ats	03/07/26	ats		
1,2-Dichlorobenzene-d4	100 %	72-127	1	03/07/26	ats	03/07/26	ats		

WET CHEMISTRY

Analysis Method: EPA 420.4 Rev. 1.0
Batch: T180375

Phenolics	<10 ug/L	10	1	03/12/26	pn	03/12/26	pn		
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QUALITY CONTROL RESULTS

Trace Project ID: 26C0249

Client Project ID: NPDES AMR.- March 2026

QC Batch: T180080

Analysis Description: Calcium, Total

QC Batch Method: EPA 200.2

Analysis Method: EPA 200.7 Rev. 4.4

METHOD BLANK: T180080-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Calcium	mg/L	<0.26	0.26	
Magnesium	mg/L	<0.050	0.050	

LABORATORY CONTROL SAMPLE: T180080-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Calcium	mg/L	16.00	15.9	100	85-115	
Magnesium	mg/L	16.00	16.3	102	85-115	

Trace Project ID: 26C0249

Client Project ID: NPDES AMR.- March 2026

QC Batch: T180080

Analysis Description: Thallium, Total

QC Batch Method: EPA 200.2

Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T180080-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Chromium	ug/L	<5.0	5.0	
Nickel	ug/L	<5.0	5.0	
Lead	ug/L	<0.55	0.55	
Antimony	ug/L	<0.25	0.25	
Selenium	ug/L	<0.50	0.50	
Thallium	ug/L	<0.38	0.38	
Zinc	ug/L	<1.2	1.2	

LABORATORY CONTROL SAMPLE: T180080-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Chromium	ug/L	50.00	52.7	105	85-115	
Nickel	ug/L	1600	1630	102	85-115	
Lead	ug/L	100.0	105	105	85-115	
Antimony	ug/L	100.0	111	111	85-115	
Selenium	ug/L	100.0	99.8	100	85-115	

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LABORATORY CONTROL SAMPLE: T180080-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Thallium	ug/L	100.0	104	104	85-115	
Zinc	ug/L	1600	1680	105	85-115	

Trace Project ID: 26C0249

Client Project ID: NPDES AMR.- March 2026

QC Batch: [CALC]

Analysis Description: Hardness (Metals)

QC Batch Method:

Analysis Method: SM 2340 B-21

Trace Project ID: 26C0249

Client Project ID: NPDES AMR.- March 2026

QC Batch: T180228

Analysis Description: 612 Chlorinated Hydrocarbons

QC Batch Method: EPA 3510C Separatory Funnel
Liquid-Liquid Extr.

Analysis Method: EPA 612

METHOD BLANK: T180228-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Hexachlorobenzene	ug/L	0.0172	0.010	BLK07
Hexachlorobutadiene	ug/L	<0.010	0.010	
Hexachlorocyclopentadiene	ug/L	<0.010	0.010	
Hexachloroethane	ug/L	<0.010	0.010	
Pentachlorophenol	ug/L	0.252	1.8	J
Tetrachloro-m-xylene (S)	%	74	18-105	

LABORATORY CONTROL SAMPLE: T180228-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Hexachlorobenzene	ug/L	0.05000	0.0428	86	26-128	
Hexachlorobutadiene	ug/L	0.05000	0.0311	62	26-107	
Hexachlorocyclopentadiene	ug/L	0.05000	0.0115	23	25-105	
Hexachloroethane	ug/L	0.05000	0.0303	61	26-87	
Pentachlorophenol	ug/L	1.800	<1.8	76	25-100	J
Tetrachloro-m-xylene (S)	%	0.06000	0.0456	76	18-105	

Trace Project ID: 26C0249

Client Project ID: NPDES AMR.- March 2026

QC Batch: T180180

Analysis Description: TTO Semi-Volatiles

QC Batch Method: EPA 625 Extraction for BNAs

Analysis Method: EPA 625.1

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METHOD BLANK: T180180-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
N-Nitrosodimethylamine	ug/L	<5.0	5.0	
Bis(2-chloroethyl)ether	ug/L	<1.0	1.0	
2-Chlorophenol	ug/L	<2.5	2.5	
Phenol	ug/L	<2.5	2.5	
1,4-Dichlorobenzene	ug/L	<2.5	2.5	
Bis(2-chloroisopropyl)ether	ug/L	<2.5	2.5	
N-Nitrosodi-n-propylamine	ug/L	<2.5	2.5	
Nitrobenzene	ug/L	<1.0	1.0	
Isophorone	ug/L	<2.5	2.5	
2-Nitrophenol	ug/L	<2.5	2.5	
2,4-Dimethylphenol	ug/L	<2.5	2.5	
Bis(2-chloroethoxy)methane	ug/L	<2.5	2.5	
1,2,4-Trichlorobenzene	ug/L	<2.5	2.5	
2,4-Dichlorophenol	ug/L	<2.5	2.5	
Naphthalene	ug/L	<2.5	2.5	
4-Chloro-3-methylphenol	ug/L	<2.5	2.5	
2,4,6-Trichlorophenol	ug/L	<1.8	1.8	
2-Chloronaphthalene	ug/L	<2.5	2.5	
Dimethyl phthalate	ug/L	<2.5	2.5	
Acenaphthylene	ug/L	<2.5	2.5	
2,6-Dinitrotoluene	ug/L	<2.5	2.5	
Acenaphthene	ug/L	<2.5	2.5	
2,4-Dinitrotoluene	ug/L	<2.5	2.5	
4-Nitrophenol	ug/L	<20	20	
2,4-Dinitrophenol	ug/L	<15	15	
Diethyl phthalate	ug/L	<2.5	2.5	
Fluorene	ug/L	<2.5	2.5	
4-Chlorophenyl phenyl ether	ug/L	<2.5	2.5	
4,6-Dinitro-2-methylphenol	ug/L	<5.0	5.0	
N-Nitrosodiphenylamine	ug/L	<2.5	2.5	
1,2-Diphenylhydrazine	ug/L	<2.5	2.5	
4-Bromophenyl phenyl ether	ug/L	<2.5	2.5	
Phenanthrene	ug/L	<1.0	1.0	
Anthracene	ug/L	<2.5	2.5	
Di-n-butyl phthalate	ug/L	<2.5	2.5	
Fluoranthene	ug/L	<1.0	1.0	
Pyrene	ug/L	<2.5	2.5	
Butyl benzyl phthalate	ug/L	<2.5	2.5	
Benzo (a) anthracene	ug/L	<1.0	1.0	
Chrysene	ug/L	<1.0	1.0	
Bis(2-ethylhexyl)phthalate	ug/L	<2.5	2.5	
Di-n-octyl phthalate	ug/L	<2.5	2.5	
Benzo (b) fluoranthene	ug/L	<1.0	1.0	
Benzo (k) fluoranthene	ug/L	<1.0	1.0	

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METHOD BLANK: T180180-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzo (a) pyrene	ug/L	<1.0	1.0	
Indeno (1,2,3-cd) pyrene	ug/L	<1.0	1.0	
Dibenz (a,h) anthracene	ug/L	<1.0	1.0	
Benzo (g,h,i) perylene	ug/L	<1.0	1.0	
2-Fluorophenol (S)	%	41	20-53	
Phenol-d5 (S)	%	29	11-40	
Nitrobenzene-d5 (S)	%	63	36-103	
2-Fluorobiphenyl (S)	%	59	36-119	
2,4,6-Tribromophenol (S)	%	61	30-105	
Terphenyl-d14 (S)	%	68	37-109	

LABORATORY CONTROL SAMPLE: T180180-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bis(2-chloroethyl)ether	ug/L	100.0	72.4	72	43-126	
2-Chlorophenol	ug/L	100.0	75.5	76	36-120	
Phenol	ug/L	100.0	32.5	33	17-120	
1,4-Dichlorobenzene	ug/L	100.0	68.0	68	39-78	
Bis(2-chloroisopropyl)ether	ug/L	100.0	80.0	80	63-139	
N-Nitrosodi-n-propylamine	ug/L	100.0	76.4	76	14-198	
Nitrobenzene	ug/L	100.0	81.4	81	54-158	
Isophorone	ug/L	100.0	58.2	58	47-180	
2-Nitrophenol	ug/L	100.0	82.3	82	45-167	
2,4-Dimethylphenol	ug/L	100.0	69.8	70	42-120	
Bis(2-chloroethoxy)methane	ug/L	100.0	79.0	79	49-165	
1,2,4-Trichlorobenzene	ug/L	100.0	68.9	69	57-130	
2,4-Dichlorophenol	ug/L	100.0	74.0	74	53-122	
Naphthalene	ug/L	100.0	73.5	73	36-120	
4-Chloro-3-methylphenol	ug/L	100.0	81.3	81	41-128	
2,4,6-Trichlorophenol	ug/L	100.0	79.9	80	52-129	
2-Chloronaphthalene	ug/L	100.0	75.7	76	65-120	
Dimethyl phthalate	ug/L	100.0	78.4	78	1-120	
Acenaphthylene	ug/L	100.0	76.3	76	54-126	
2,6-Dinitrotoluene	ug/L	100.0	81.4	81	68-137	
Acenaphthene	ug/L	100.0	78.1	78	60-132	
2,4-Dinitrotoluene	ug/L	100.0	84.5	84	48-127	
4-Nitrophenol	ug/L	100.0	35.4	35	13-129	
2,4-Dinitrophenol	ug/L	100.0	79.0	79	1-173	
Diethyl phthalate	ug/L	100.0	79.8	80	1-120	
Fluorene	ug/L	100.0	80.9	81	70-120	
4-Chlorophenyl phenyl ether	ug/L	100.0	76.4	76	38-145	

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LABORATORY CONTROL SAMPLE: T180180-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
4,6-Dinitro-2-methylphenol	ug/L	100.0	84.2	84	53-130	
1,2-Diphenylhydrazine	ug/L	100.0	75.1	75	55-115	
4-Bromophenyl phenyl ether	ug/L	100.0	72.8	73	65-120	
Phenanthrene	ug/L	100.0	78.8	79	65-120	
Anthracene	ug/L	100.0	80.3	80	43-120	
Di-n-butyl phthalate	ug/L	100.0	77.0	77	8-120	
Fluoranthene	ug/L	100.0	69.8	70	43-121	
Pyrene	ug/L	100.0	70.7	71	70-120	
Butyl benzyl phthalate	ug/L	100.0	76.4	76	1-140	
Benzo (a) anthracene	ug/L	100.0	84.2	84	42-133	
Chrysene	ug/L	100.0	80.1	80	44-140	
Bis(2-ethylhexyl)phthalate	ug/L	100.0	75.8	76	29-137	
Di-n-octyl phthalate	ug/L	100.0	88.1	88	19-132	
Benzo (b) fluoranthene	ug/L	100.0	80.6	81	42-140	
Benzo (k) fluoranthene	ug/L	100.0	76.9	77	25-146	
Benzo (a) pyrene	ug/L	100.0	79.6	80	32-148	
Indeno (1,2,3-cd) pyrene	ug/L	100.0	88.1	88	1-151	
Dibenz (a,h) anthracene	ug/L	100.0	81.8	82	1-200	
Benzo (g,h,i) perylene	ug/L	100.0	111	111	1-195	
2-Fluorophenol (S)	%	100.0	53.2	53	20-53	
Phenol-d5 (S)	%	100.0	35.7	36	11-40	
Nitrobenzene-d5 (S)	%	100.0	82.2	82	36-103	
2-Fluorobiphenyl (S)	%	100.0	77.5	77	36-119	
2,4,6-Tribromophenol (S)	%	100.0	84.2	84	30-105	
Terphenyl-d14 (S)	%	100.0	78.1	78	37-109	

MATRIX SPIKE: T180180-MS1

Original: 26C0249-01

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
Bis(2-chloroethyl)ether	ug/L	0	104.0	66.0	64	12-158	
2-Chlorophenol	ug/L	0	104.0	70.3	68	23-134	
Phenol	ug/L	0	104.0	31.1	30	5-120	
1,4-Dichlorobenzene	ug/L	0	104.0	63.7	61	31-83	
Bis(2-chloroisopropyl)ether	ug/L	0	104.0	75.8	73	36-166	
N-Nitrosodi-n-propylamine	ug/L	0	104.0	65.8	63	1-230	
Nitrobenzene	ug/L	0	104.0	68.1	65	35-180	
Isophorone	ug/L	0	104.0	51.1	49	21-196	
2-Nitrophenol	ug/L	0	104.0	72.7	70	29-182	
2,4-Dimethylphenol	ug/L	0	104.0	48.5	47	32-120	
Bis(2-chloroethoxy)methane	ug/L	0	104.0	67.3	65	33-184	
1,2,4-Trichlorobenzene	ug/L	0	104.0	61.4	59	44-142	

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MATRIX SPIKE: T180180-MS1

Original: **26C0249-01**

Parameter	Units	Original Result	Spike Conc.	MS Result	MS % Rec	% Rec Unit	Notes
2,4-Dichlorophenol	ug/L	0	104.0	73.6	71	39-135	
Naphthalene	ug/L	0	104.0	65.8	63	21-133	
4-Chloro-3-methylphenol	ug/L	0	104.0	79.5	76	22-147	
2,4,6-Trichlorophenol	ug/L	0	104.0	81.1	78	37-144	
2-Chloronaphthalene	ug/L	0	104.0	71.2	69	60-120	
Dimethyl phthalate	ug/L	0	104.0	74.0	71	1-120	
Acenaphthylene	ug/L	0	104.0	73.0	70	33-145	
2,6-Dinitrotoluene	ug/L	0	104.0	79.0	76	50-158	
Acenaphthene	ug/L	0	104.0	76.1	73	47-145	
2,4-Dinitrotoluene	ug/L	0	104.0	82.8	80	39-139	
4-Nitrophenol	ug/L	0	104.0	42.6	41	1-132	
2,4-Dinitrophenol	ug/L	0	104.0	89.2	86	1-191	
Diethyl phthalate	ug/L	0	104.0	77.5	75	1-120	
Fluorene	ug/L	0	104.0	80.1	77	59-121	
4-Chlorophenyl phenyl ether	ug/L	0	104.0	73.9	71	25-158	
4,6-Dinitro-2-methylphenol	ug/L	0	104.0	85.8	83	1-181	
1,2-Diphenylhydrazine	ug/L	0	104.0	77.1	74	55-115	
4-Bromophenyl phenyl ether	ug/L	0	104.0	72.2	69	53-127	
Phenanthrene	ug/L	0	104.0	79.8	77	54-120	
Anthracene	ug/L	0	104.0	77.8	75	50-107	
Di-n-butyl phthalate	ug/L	0	104.0	77.2	74	1-120	
Fluoranthene	ug/L	0	104.0	71.9	69	26-137	
Pyrene	ug/L	0	104.0	71.5	69	52-120	
Butyl benzyl phthalate	ug/L	0	104.0	78.8	76	1-152	
Benzo (a) anthracene	ug/L	0	104.0	82.0	79	33-143	
Chrysene	ug/L	0	104.0	77.6	75	17-168	
Bis(2-ethylhexyl)phthalate	ug/L	0	104.0	75.7	73	8-158	
Di-n-octyl phthalate	ug/L	0	104.0	86.0	83	4-146	
Benzo (b) fluoranthene	ug/L	0	104.0	80.0	77	24-159	
Benzo (k) fluoranthene	ug/L	0	104.0	81.3	78	11-162	
Benzo (a) pyrene	ug/L	0	104.0	80.4	77	17-163	
Indeno (1,2,3-cd) pyrene	ug/L	0	104.0	83.3	80	1-171	
Dibenz (a,h) anthracene	ug/L	0	104.0	79.8	77	1-227	
Benzo (g,h,i) perylene	ug/L	0	104.0	84.0	81	1-219	
2-Fluorophenol (S)	%		104.0	43.8	42	20-53	
Phenol-d5 (S)	%		104.0	31.2	30	11-40	
Nitrobenzene-d5 (S)	%		104.0	71.9	69	36-103	
2-Fluorobiphenyl (S)	%		104.0	71.2	69	36-119	
2,4,6-Tribromophenol (S)	%		104.0	86.7	83	30-105	
Terphenyl-d14 (S)	%		104.0	52.9	51	37-109	

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Trace Project ID: 26C0249
Client Project ID: NPDES AMR.- March 2026

QC Batch: T180116	Analysis Description: TTO Volatiles
QC Batch Method: EPA 624 Purge and Trap	Analysis Method: EPA 624.1

METHOD BLANK: T180116-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Chloromethane	ug/L	<1.0	1.0	
Vinyl chloride	ug/L	<1.0	1.0	
Bromomethane	ug/L	<5.0	5.0	
Chloroethane	ug/L	<2.0	2.0	
Acrolein	ug/L	<10	10	
1,1-Dichloroethene	ug/L	<1.0	1.0	
Methylene chloride	ug/L	<5.0	5.0	
Acrylonitrile	ug/L	<2.0	2.0	
trans-1,2-Dichloroethene	ug/L	<1.0	1.0	
1,1-Dichloroethane	ug/L	<1.0	1.0	
Chloroform	ug/L	<1.0	1.0	
1,1,1-Trichloroethane	ug/L	<1.0	1.0	
Carbon tetrachloride	ug/L	<1.0	1.0	
Benzene	ug/L	<1.0	1.0	
1,2-Dichloroethane	ug/L	<1.0	1.0	
Trichloroethene	ug/L	<1.0	1.0	
1,2-Dichloropropane	ug/L	<1.0	1.0	
cis-1,2-Dichloroethene	ug/L	<1.0	1.0	
Bromodichloromethane	ug/L	<1.0	1.0	
2-Chloroethylvinyl ether	ug/L	<5.0	5.0	
cis-1,3-Dichloropropene	ug/L	<1.0	1.0	
Toluene	ug/L	<1.0	1.0	
trans-1,3-Dichloropropene	ug/L	<1.0	1.0	
1,1,2-Trichloroethane	ug/L	<1.0	1.0	
Tetrachloroethene	ug/L	<1.0	1.0	
Dibromochloromethane	ug/L	<1.0	1.0	
Chlorobenzene	ug/L	<1.0	1.0	
Ethylbenzene	ug/L	<1.0	1.0	
Bromoform	ug/L	<1.0	1.0	
1,1,2,2-Tetrachloroethane	ug/L	<1.0	1.0	
1,3-Dichlorobenzene	ug/L	<1.0	1.0	
1,4-Dichlorobenzene	ug/L	<1.0	1.0	
1,2-Dichlorobenzene	ug/L	<1.0	1.0	
1,2-Dichloroethane-d4 (S)	%	98	68-133	
Toluene-d8 (S)	%	102	75-120	
4-Bromofluorobenzene (S)	%	103	69-119	
1,2-Dichlorobenzene-d4 (S)	%	101	72-127	

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LABORATORY CONTROL SAMPLE: T180116-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Chloromethane	ug/L	50.00	47.5	95	1-205	
Vinyl chloride	ug/L	50.00	45.9	92	5-195	
Bromomethane	ug/L	50.00	48.4	97	15-185	
Chloroethane	ug/L	50.00	56.6	113	40-160	
Acrolein	ug/L	50.00	76.7	153	60-140	BS05
1,1-Dichloroethene	ug/L	50.00	54.6	109	50-150	
Methylene chloride	ug/L	50.00	51.5	103	60-140	
Acrylonitrile	ug/L	50.00	73.1	146	60-140	BS05
trans-1,2-Dichloroethene	ug/L	50.00	52.3	105	70-130	
1,1-Dichloroethane	ug/L	50.00	57.1	114	70-130	
Chloroform	ug/L	50.00	51.8	104	70-135	
1,1,1-Trichloroethane	ug/L	50.00	52.8	106	70-130	
Carbon tetrachloride	ug/L	50.00	53.2	106	70-130	
Benzene	ug/L	50.00	55.7	111	65-135	
1,2-Dichloroethane	ug/L	50.00	51.8	104	70-130	
Trichloroethene	ug/L	50.00	49.7	99	65-135	
1,2-Dichloropropane	ug/L	50.00	54.2	108	35-165	
cis-1,2-Dichloroethene	ug/L	50.00	53.3	107	70-130	
Bromodichloromethane	ug/L	50.00	51.5	103	65-135	
2-Chloroethylvinyl ether	ug/L	50.00	50.6	101	1-225	
cis-1,3-Dichloropropene	ug/L	50.00	51.4	103	25-175	
Toluene	ug/L	50.00	54.7	109	70-130	
trans-1,3-Dichloropropene	ug/L	50.00	51.8	104	50-150	
1,1,2-Trichloroethane	ug/L	50.00	54.7	109	70-130	
Tetrachloroethene	ug/L	50.00	53.0	106	70-130	
Dibromochloromethane	ug/L	50.00	51.4	103	70-135	
Chlorobenzene	ug/L	50.00	56.4	113	65-135	
Ethylbenzene	ug/L	50.00	51.9	104	60-140	
Bromoform	ug/L	50.00	53.8	108	70-130	
1,1,2,2-Tetrachloroethane	ug/L	50.00	56.3	113	60-140	
1,3-Dichlorobenzene	ug/L	50.00	54.4	109	70-130	
1,4-Dichlorobenzene	ug/L	50.00	54.8	110	65-135	
1,2-Dichlorobenzene	ug/L	50.00	52.8	106	65-135	
1,2-Dichloroethane-d4 (S)	%	30.00	29.3	98	68-133	
Toluene-d8 (S)	%	30.00	30.8	103	75-120	
4-Bromofluorobenzene (S)	%	30.00	29.8	99	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.00	30.0	100	72-127	

Trace Project ID: 26C0249

Client Project ID: NPDES AMR.- March 2026

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QC Batch: T180375

Analysis Description: Phenols, Total

QC Batch Method: EPA 420.4 Rev. 1.0

Analysis Method: EPA 420.4 Rev. 1.0

METHOD BLANK: T180375-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenolics	ug/L	<10	10	

LABORATORY CONTROL SAMPLE: T180375-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenolics	ug/L	100.0	87.0	87	73-110	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

Definitions and Notes

Trace ID: 26C0249-01

Analysis: EPA 625.1

Benzidine	Note MS07 : The MS recovery was out of control.
------------------	---

Trace ID: T180116-BS1

Analysis: EPA 624.1

Acrolein	Note BS05 : The BS recovery was out of control high. Because there were no positive results for this analyte in this QC batch, no data require qualification.
Acrylonitrile	Note BS05 : The BS recovery was out of control high. Because there were no positive results for this analyte in this QC batch, no data require qualification.

Trace ID: T180228-BLK1

Analysis: EPA 612

Hexachlorobenzene	Note BLK07 : A positive result for this analyte was found in the method blank. Because no positive result was found in the sample, no data require qualification.
--------------------------	---

Trace ID: T180228-BSD1

Analysis: EPA 612

Hexachlorocyclopentadiene	Note BS07 : One of the following: BS, BSD or RPD was out of control. Because two of the results are in control, no data requires qualification.
----------------------------------	---

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26C0249

Cadillac, City of

Project Manager: Tim Brewer

Sample Log In Checklist

Date: 3/5/26	Original Observation	Corrected Temperature	IR-9 (CF: +0.2°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: -0.8°C)	IR-14 (CF: -1.2°C)	SR1 (CF: -0.3°C)	SR2 (CF: -0.2°C)	Temp Blank	Client Sample
Time: 1648										
Initials: SAB										
Package Description: Cooler										
Package Temp °C	-0.8	-1.6								
Representative Sample Temp °C	2.2	2.0								

Sample Receipt

Yes No

- ☒ ☐ Received on ice or other coolant
☒ ☐ Ice still present upon receipt
☐ ☒ Custody seals present
☒ Trace Courier ☐ Client Drop-off
- ☐ Yes ☐ No Custody seals intact (if applicable)
☐ UPS ☐ Fed Ex ☐ US Mail ☐ Other

Sample Condition

Yes No N/A

- ☒ ☐ ☐ All sample containers arrived unbroken and labeled
☒ ☐ ☐ Sufficient sample to run requested analyses
☒ ☐ ☐ Correct chemical preservative added to samples
☐ ☒ ☐ Samples preserved at Trace (if yes, fill in preservation information below)
Preservation - Reagent: _____ LIMS ID: _____ Date/Time: _____
☒ ☐ ☐ Chemical preservation verified, check EMD pH test strip used (if applicable)
☒ pH 0-2.5 (Lot: HC461264) ☐ pH 11.0-13.0 (Lot: HC022540) ☐ Other
☐ ☒ ☐ Air bubbles absent from VOAs (no bubble greater than 6mm) 26 mm bubble in one VOA

Chain of Custody (COC)

Yes No

- ☒ ☐ All bottle labels agree with COC
☒ ☐ COC filled out properly
☒ ☐ COC signed by client

Notes:

CERTIFICATE OF ANALYSIS

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